

Faiyaz Rahman

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4+ years delivering full-stack systems across AI, infra, and product. Architected multi-cloud fintech platforms (AWS EU / Alibaba China), scaled Kafka to 1.5M msg/day, built Bedrock/RAG pipelines for quantitative reporting. Comfortable in NestJS/TypeScript APIs, Vue/React UIs, and the GitOps glue between them.

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, Scala, Kotlin, Go, SQL, Terraform, Helm

AI/ML: AWS Bedrock, RAG, Vector DBs, Prompt Engineering

Frameworks: Vue.js, React, Nest.js, Spring, Ktor, Flask, FastAPI, Pandas

Platforms: Docker, Kubernetes, ArgoCD, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch

Clouds: AWS (EC2, S3, Lambda, EKS, RDS), Alibaba Cloud

Architecture: Microservices, Event-Driven, CQRS, GitOps, CI/CD, REST

EXPERIENCE

Full Stack Developer

Apr 2025 – present

Premialab

Full-time

- **Hybrid Search Screener:** Built a sub-300ms hybrid search screener over 100k+ instruments — evaluated Qdrant prototype vs pgvector/HNSW, chose Postgres for operational simplicity; fused Titan dense embeddings + BM25 sparse via RRF; ag-Grid frontend with virtualized rows, multi-tenant filter scopes, saved views. aggregated metrics, P95 < 500ms cold, < 300ms warm.
- **AI-Driven No-Code Reporting:** Architected a modular CRUD persistence engine using Vue.js, FastAPI, NestJS, and AWS Bedrock (Claude 3.5 Sonnet); implemented a RAG pipeline over financial reports to stream automated commentary alongside 20+ analytics components with exact PDF layout parity.
- **Factor Comparison & Peer Group Analytics:** Engineered end-to-end peer group workflows and Premialab factor comparison engines using Python (Pandas/ NumPy/ Jupyter) and PostgreSQL to compute volume-adjusted betas, rolling regressor graphs, and KDE distributions; built an automated regression layer to accelerate feature onboarding.
- **Market Regime Predictive Modeling:** Engineered a low-latency, end-to-end quantitative data pipeline integrating a high-performance C++ calculation engine containerized via Docker; mapped live macro signals directly to downstream modules using Threshold methodologies to segment portfolio performances across several market regimes.
- **Client-Side Report Automation:** Eliminated a flaky, legacy headless browser infrastructure by architecture-shifting to a client-side jsPDF implementation; removed unpredictable server-side rendering failures while achieving an 80% reduction in report generation time and ensuring exact WYSIWYG parity.
- **Portfolio Simulation Automated Testing:** Engineered a TypeScript pipeline that parses financial analyst Excel models to dynamically generate Jest integration tests, validating nested performance attribution metrics; enforced strict mathematical parity across varying benchmarks, active return calculations, and diverse rebalancing strategies, trapping deep calculation regressions instantly before deployment.
- **Enterprise Whitelabel Enablement:** Enabled tier-one institutional rollouts for global clients including Société Générale, Royal Bank of Canada, and Schrodgers by building robust tenant whitelabel configurations; automated team onboarding workflows, including third-party data subscriptions and default portfolio provisioning, to shorten client time-to-value.
- **Developer Experience (DX) Overhaul:** Accelerated local and CI workflow speeds by migrating legacy ESLint configs to Rust-based Oxlint; integrated SonarQube to establish an automated code review safety net while engineering custom Jira-to-PR automation tools to eliminate manual developer overhead.

Junior Full Stack Developer

Jul 2023 – Feb 2025

KM.ON Asia Limited

Full-time

- **Multi-Cloud Data Residency:** Spearheaded the end-to-end architecture, development, and deployment of a full-stack React.js and Express.js energy platform ([news](#)) across AWS (EU) and Alibaba Cloud (China); owned the full lifecycle of Terraform IaC, ArgoCD, and Helm charts while independently engineering custom data-routing layers within the application to enforce strict GDPR and PIPL compliance.
- **Terabyte-Scale Aggregation:** Engineered a two-tier fan-out Python/ Lambda/ FCompute data pipeline to ingest over 1.5 TB of raw IoT log histories from S3/OSS into active relational storage; implemented memory-capped streaming mechanics monitored via Grafana/CloudWatch to eliminate OOM failures.
- **Time-Series Database Scaling:** Deployed and maintained an AWS RDS Aurora PostgreSQL cluster extended with TimescaleDB hypertables partitioned by telemetry timestamps; handled a sustained write-throughput of 2,000+ operations per second while preventing index bloat.

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- **Zero-Downtime Event Streaming:** Executed a live production infrastructure migration in **Java 17** from legacy messaging layers to **Strimzi Kafka** and **Karapace**; authored custom **KafkaConnectors** processing **10M+ messages daily** with inline sink verification and Dead Letter Queues (DLQ) to ensure 100% data parity.
- **High-Throughput IoT Network:** Developed a **Scala** microservice running over **MQTT** to stream bidirectional binary BLOB configurations between APIs and textile machinery; built an edge command hub with custom exponential back-off loops to guarantee delivery of proprietary KMSF payloads over unstable networks.
- **Security Supply Chain & GitOps:** Integrated programmatic **SBOM** generation and **Trivy** vulnerability scans into the CI/CD pipeline for 15+ microservices to enforce strict EU/ CN compliance; configured automated semantic versioning workflows that parsed Git commit hooks to dynamically tag and build Docker artifacts.

Software Engineer

Jun 2022 - Sep 2022

Reap Technologies

Internship

- **Monolith Refactoring & TDD:** Migrated legacy monolithic codebase into Controller-Service-Repository (CSR) architecture using Test-Driven Development (TDD) principles; engineered robust integration test suites with Docker scaffolding and mocked services to eliminate test flakiness and pave way for microservices transition.

IT Strategist

Feb 2021 – Aug 2021

BNP Paribas CIB

Contract

- **Latency-Aware Sync Engine:** Engineered a distributed, fan-out data replication engine in **Python** configured to dynamically route payloads based on live server latencies and node clustering; bypassed GIL limitations via multiprocessing to synchronize terabyte-scale log data across **200+ nodes**.
- **Real-Time Analytics Pipeline:** Accelerated data delivery by designing an ingestion pipeline using **Kafka** and **Elasticsearch**; optimized cluster indexing strategies to achieve sub-second dashboard refreshes for high-cardinality trading logs.
- **Compliant Deployment Orchestration:** Developed automated shell scripts and platform workflows to enforce the mandatory **Four-Eyes Principle**; configured multi-party approval validation for staging-to-production deployments and integrated programmatic **systemd** health checks.
- **High-Throughput Packet Replay Engine:** Optimized a multi-threaded **C++** simulation system replaying network packets from **.pcap histories** into trading systems; resolved thread-synchronization bottlenecks by replacing blocking request-count evaluation loops with non-blocking condition variables.

PROJECTS

Homelab

2025 – present

20+ self-hosted services with Golang deployment tooling and Tailscale networking.

- Built a **Golang** CLI toolchain (**uv**, **rich**) around a centralized service registry that drives the full lifecycle of 20+ services — **age**-encrypted secrets, prioritized **Docker Compose** orchestration, HTTP health probing, and wildcard TLS management — consolidating from earlier bash/ TOML config layer into single source of truth.
- Designed a zero-exposure network architecture by migrating from WireGuard + Duck DNS + dnsmasq to **Tailscale** mesh VPN with MagicDNS, **Pi-hole** authoritative wildcard DNS, and **Caddy** reverse proxy with a self-managed two-tier PKI — all services remotely accessible without a single forwarded port.
- Deployed a multi-layer observability stack: **VictoriaMetrics** (migrated from Prometheus for resource efficiency), **Loki** + **Promtail** for log aggregation, **Grafana** dashboards, **Alertmanager**, **Gatus** uptime probing (replacing Uptime Kuma), and **Scrutiny** for S.M.A.R.T. disk health.
- Enforced engineering rigor via **ruff**, **mypy**, and **pytest** pre-commit hooks, SHA256-pinned images, and append-only changelog auditing; Built to deeply understand observability and zero-trust networking outside managed cloud.

EDUCATION

The Chinese University of Hong Kong

Sep 2018 – Jun 2023

Bachelor of Science in Computer Science

Second Class Honours Upper Division (2:1)

- **Concentration:** Database and Information Systems
- **Honors:** ‘Full Ride’ Admission Scholarship, The ‘Sunny’ Award
- **Activities:** English Debating Team, Google Developers Student Club